This booklet is part of a continuing patient education series on the thyroid gland and thyroid disease. Additional booklets available in this series are:

Getting to Know Your Medication
How the Thyroid Gland Can Affect Your Weight
Hyperthyroidism and Your Health
Hypothyroidism and Your Health
The Postpartum Period and Your Thyroid
The Thyroid Gland and Your Health
Thyroid Disease and Diabetes
Thyroid Disease in Women
Understanding Goiter
Understanding Graves' Disease
Understanding Thyroid Cancer
Understanding Thyroid Nodules
Understanding TSH Testing
Understanding Your Child's Thyroid
What Is Hashimoto's Disease?

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Thyroid Disease and Aging

Thyroid Disease and Aging—What You Need to Know

The chance of developing a thyroid disorder changes as you age. For example, while about 4% to 8% of the general population may have mild thyroid failure (a condition in which the thyroid gland begins to work improperly), 9% to 16% of people age 60 and over may have the disorder.

Luckily, most thyroid disorders are treatable, and the rate of treatment success is high.

The following pages contain basic information about the thyroid gland, its role in the body, and thyroid disorders. **In addition, this booklet will discuss the issues surrounding thyroid disease in the older patient.**

Remember, no single booklet can answer all of your questions or replace speaking with doctors, nurses, and other members of the healthcare team. If you have any questions or concerns about the thyroid or thyroid disorders, please check with your doctor.

Notes:

For more information on the thyroid gland, please contact:

The Thyroid Foundation of America
Ruth Sleeper Hall, RSL 350
40 Parkman Street
Boston, MA 02114-2698
(send self-addressed, stamped envelope)

The Thyroid Society for Education & Research
7515 S. Main Street, Suite 545
Houston, TX 77030
1-800-THYROID

National Graves' Disease Foundation
2 Tsitsi Court
Brevard, NC 28712

Or visit our web site at:
www.glandcentral.com

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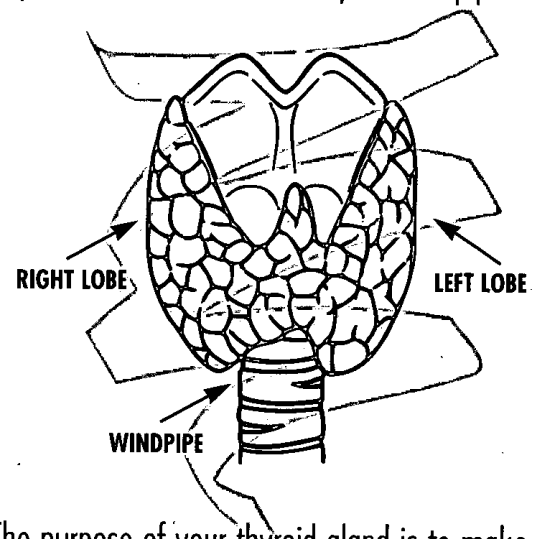
Your Thyroid— The Basics

Your thyroid is a gland located at the base of your neck, just below your Adam's apple. It is shaped like a butterfly – each "wing," or **lobe**, of your thyroid lies on either side of your windpipe.



The purpose of your thyroid gland is to make, store, and release **thyroid hormone** into your blood. Thyroid hormone affects essentially every cell in your body, and helps control your body's functions. If you have too little thyroid hormone in your blood, your body slows down. This is a condition called **hypothyroidism**. If you have too much thyroid hormone in your blood, your body speeds up. This is a condition called **hyperthyroidism**.

A Final Note

With increasing age, it is more likely that you may be taking medications for other conditions, such as heart disease or high cholesterol. If you think you may have a thyroid disorder, or you are currently being treated for one, make sure you tell your doctor about every medication you take, both prescription and over-the-counter. You may be taking medications that are no longer necessary once your thyroid condition is treated. Moreover, other medications may have an effect on your thyroid treatment or be affected by your treatment and may need to be adjusted. Your doctor should be told so that he or she can design your thyroid treatment around your other therapies. Even things like iron supplements and food made with soybean flour can interfere with the success of your treatment, so be sure to keep your doctor informed about your health habits.

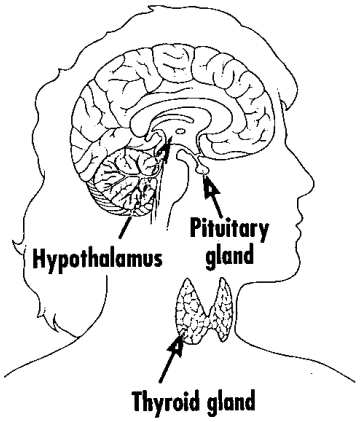
Hypothyroidism can result in high cholesterol levels, slowed heartbeat, edema, and depression—conditions that can be serious for the older patient. If you have hypothyroidism and are not being treated, or if you are being given too low a dose of thyroid hormone for the treatment of hypothyroidism, you may be at risk for these problems.

On the other hand, if you have too much thyroid hormone in your body due to untreated hyperthyroidism, inadequately treated hyperthyroidism, or overtreated hypothyroidism (overreplacement of thyroid hormone), you may be at risk for an increased heart rate, irregular pulse, heart failure, and reduced bone density, which can lead to osteoporosis. Since the older patient is particularly vulnerable to these conditions, it is especially important to treat them right away.

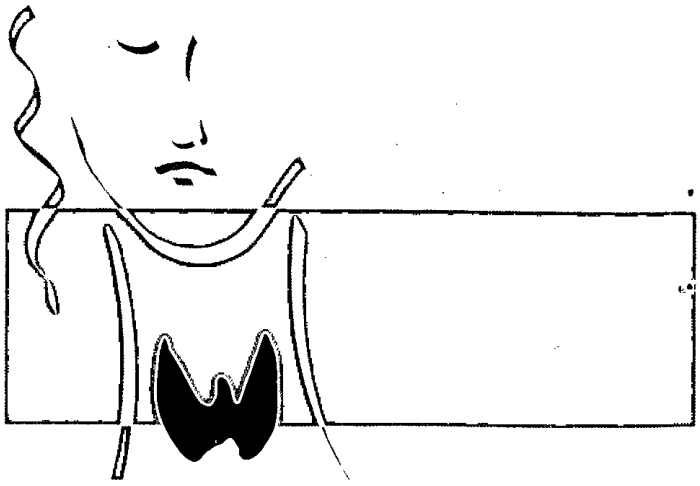
The amount of thyroid hormone made by your thyroid gland is adjusted by another gland called the **pituitary**, as well as a part of your brain called the **hypothalamus**.

- The pituitary gland makes hormones that control many other glands in your body
- The hypothalamus helps the pituitary control many of your glands, as well as helping to control other bodily functions, such as thirst, hunger, sleep, and body temperature

The thyroid gland, pituitary gland, and hypothalamus all work together to control the amount of thyroid hormone in your body. With the pituitary controlling most of the action, these structures work in a way that is similar to the way a thermostat controls the temperature in a room.



For instance, just as the thermometer in the thermostat senses the temperature of a room, your **pituitary** constantly senses the amount of thyroid hormone in your blood. If there is not enough thyroid hormone, your pituitary senses the need to "turn on the heat." It does this by releasing more **Thyroid-Stimulating Hormone (TSH)**, which signals your thyroid to make more thyroid hormone. Your thyroid gland then makes and releases the hormone directly into your bloodstream and restores the amount of thyroid hormone in your blood to normal.



Doing Your Part

No matter what the disorder or the treatment course, you play a vital role in maintaining your health. By following your doctor's instructions and reporting any symptoms that you experience to your doctor right away, you can prevent some very serious health problems that can occur.

In particular, it is extremely important that you take your medication exactly as directed by your doctor. Over time, the impact of too much or too little thyroid hormone in your body can be very serious.

Thyroid Nodules and Cancer

Thyroid cancer is uncommon, accounting for little more than 1% of all cancers in the US. Furthermore, the outcome with treatment is usually excellent—most thyroid cancers can be totally removed with surgery and a complete cure is common.

The same procedures for the evaluation and treatment of thyroid nodules and cancer in younger patients are followed for older patients.

To get a better idea of the procedures and alternatives involved, pick up a copy of *Understanding Thyroid Nodules* or *Understanding Thyroid Cancer.*

Your pituitary then senses that there is just the right amount of thyroid hormone in your body. With your thyroid hormone levels now restored to normal, your pituitary slows its production of TSH back down to normal.

Thyroid Disorders—A Brief Look at What Can Go Wrong

Hypothyroidism

If your thyroid gland produces too little thyroid hormone, you may have a condition called **hypothyroidism**. If you are hypothyroid, you also have a high level of TSH (which is your body's signal to "turn on the heat," or to step up the production and release of thyroid hormone). Having too little thyroid hormone causes your body to slow down. As a result, you may feel tired, have a slow heartbeat, or feel depressed, besides having other symptoms.

Mild Thyroid Failure

A more mild form of hypothyroidism is **mild thyroid failure**. In this case, your thyroid hormone levels are normal, but you have a high TSH level. This may mean that TSH is making your thyroid gland work extra hard just to maintain normal levels of hormone in your blood. This condition often leads to hypothyroidism. People with mild thyroid failure generally don't show any obvious symptoms, which makes it very difficult for doctors to identify without performing a TSH test.

Hypothyroidism

Just as with hyperthyroidism, your doctor will have special concerns for the older patient with hypothyroidism. Although the treatment is the same for patients of all ages—thyroid hormone replacement therapy—your body's need for thyroid hormone decreases as you age. As a result, your doctor will probably start you on a lower dosage than a younger patient, and may increase your dose more slowly until the right dose of thyroid hormone is found for you. Patients with heart problems also require special care with regard to treatment of hypothyroidism.

Once your doctor has adjusted your dose of thyroid hormone so that you are getting the right amount, he or she will prescribe that dose of medication for you and monitor your blood levels periodically.

Special Considerations

Hyperthyroidism

If you are diagnosed with hyperthyroidism, your doctor will probably treat you with **radioactive iodine** therapy. The iodine goes directly to your thyroid gland and destroys thyroid cells, thereby decreasing the production of thyroid hormone.

Other options for treatment include **surgery** and **antithyroid drugs**. Because of the trauma involved, surgery is usually not chosen for older hyperthyroid patients. Antithyroid drugs may be given to you for a period of time before radioactive iodine therapy to slow down your thyroid gland and lower thyroid hormone levels in your blood.

Patients with heart problems require special care with regard to hyperthyroidism and its treatment. Therefore, if you have any heart problems, your doctor will need to know.

Hyperthyroidism

If your thyroid gland produces too much thyroid hormone, you may have a condition known as **hyperthyroidism**. If you are hyperthyroid, you also have a low level of TSH in your blood because your pituitary has decreased its TSH signals to your thyroid gland. Having too much thyroid hormone causes your body to speed up, resulting in nervousness, fast heartbeat, and unexplained weight loss, among other symptoms.

Mild (Subclinical) Hyperthyroidism

If your thyroid hormone levels are normal, but you have a low TSH level, you may have a condition known as **mild (subclinical) hyperthyroidism**. This often progresses to hyperthyroidism at a later point, and should be monitored carefully by your doctor.

Goiter

A **goiter** simply is an enlargement of the thyroid gland. Your thyroid may be enlarged because it's not working properly, as in hypothyroidism and hyperthyroidism. A goiter may also result from thyroid nodules or an inflammation of the thyroid gland.

Thyroid Nodules

Thyroid nodules, or **lumps** in the thyroid gland, are more likely to be present in older patients. Most nodules are relatively harmless and may not even need to be treated, but your doctor will need to rule out the possibility that the nodule is either cancerous or causing hyperthyroidism.

Your Laboratory Results

- Your doctor will likely order blood tests to find out how your thyroid gland is functioning. The best single test is the **TSH test**. If your TSH level is outside the normal range, it is likely that you have a thyroid disorder

- Other tests that your doctor may order are:

—**Thyroid hormone tests**. These are direct measurements of the amount of thyroid hormone in your blood

—**Thyroid antibody test**. This test will help your doctor find out if your immune system is affecting your thyroid function

—**Radioactive iodine uptake test**. Radioactive iodine is a substance that you swallow in tablet form. Between 6 to 24 hours later, your doctor uses a special piece of equipment to measure how much of the radioactive iodine your thyroid absorbed. You may also have an image taken of your thyroid gland called a **thyroid scan**. Your doctor can use this information as a guide toward a correct diagnosis

Your Physical Condition

- If a thyroid disorder is suspected, your doctor will probably give you a physical exam to look for signs and symptoms

- Also, your doctor will look for an enlarged thyroid, or **goiter**. Your doctor will examine your neck, pressing on, or **palpating**, the area where your thyroid is located to determine if a goiter is present

- Your doctor will also examine your thyroid gland to determine if there are thyroid nodules present. Thyroid nodules are lumps in the thyroid gland that can be made up of a group of thyroid cells or fluid-filled cysts. Sometimes, but very rarely, they can be cancerous

- In addition, your weight, blood pressure, and pulse will be measured

- Your doctor may examine your eyes, skin, heart, and nervous system, as well. All of these systems are affected by hyperthyroidism

Thyroid Disorders in the Older Patient

The risk of having a thyroid disorder generally increases as you age. Hypothyroidism, the most common thyroid disorder, occurs more frequently in older individuals. For example, while about 4% to 8% of the general population may have early stages of hypothyroidism, or **mild thyroid failure**, 9% to 16% of people age 60 and over are believed to have mild thyroid failure.

Health disorders are usually accompanied by signs and symptoms that let the patient know something is wrong, and help the doctor pinpoint the problem. However, thyroid disorders, especially hypothyroidism, are often difficult to identify because their signs and symptoms are very subtle, and seem like common problems related to getting older (for example, dry skin, constipation, changes in weight, muscle and joint pain). Moreover, older patients may experience *fewer* symptoms than younger patients, making it even harder for doctors to discover thyroid disorders.

Nevertheless, the following list will give you an idea of the variety of signs and symptoms to be aware of. While some signs and symptoms are more common in older patients, any of the signs listed can be experienced at any age.

Signs & Symptoms of Hypothyroidism

- Tiredness
- Weight gain
- Dry skin
- Often feeling cold
- Coarse, dry hair or hair loss
- Hoarseness
- Constipation
- Yellowish skin
- Enlarged thyroid gland (goiter)

Those more likely than others to become hyperthyroid...

- ✔ are women, 30 years or older
- ✔ are pregnant or have given birth in the last 6 months*
- ✔ have other autoimmune conditions, such as:
  - —Addison's disease
  - —Type I diabetes
  - —Pernicious anemia
  - —Rheumatoid arthritis
  - —Systemic lupus erythematosus (SLE or lupus)

*__Note:__ If you've had a baby in the past 6 months, you may be at a higher risk for developing hyperthyroidism due to autoimmune thyroid disease.

Your Likelihood of Developing a Disorder

Some individuals are more likely than others to become hypothyroid. These include:

- ✔ women, especially those over 40
- ✔ postpartum women (within 6 months after delivery)
- ✔ people with high cholesterol
- ✔ people experiencing depression
- ✔ people who have had thyroid disease before
- ✔ people with a family history of autoimmune thyroid disease
- ✔ people who have been treated for *hyper*thyroidism
- ✔ people who have had x-ray therapy of the head or neck for cancers or Hodgkin's disease
- ✔ people who have taken or who are taking lithium or amiodarone
- ✔ people with nonthyroid autoimmune disease, for example:
  - —Addison's disease
  - —Type I diabetes
  - —Pernicious anemia
  - —Rheumatoid arthritis
  - —Systemic lupus erythematosus (SLE or lupus)

- Difficulty remembering things
- Confusion
- Decreased concentration
- Depression
- Withdrawn behavior
- Irregular or heavy menstruation
- Infertility
- Muscle aches
- Lack of coordination
- High cholesterol
- Slowed heartbeat
- Low body temperature

Signs & Symptoms of Hyperthyroidism

- Nervousness and irritability
- Fast, irregular heartbeat
- Heat intolerance and increased sweating
- Shaky hands (tremor)
- Weight loss
- Changes in appetite (usually increased appetite)
- More frequent bowel movements
- Tiredness and muscle weakness
- Thyroid enlargement (goiter)
- Sleep disturbances (for example, insomnia)
- Shortness of breath, especially with exercise
- Impaired fertility
- Eye irritation, double vision, bulging eyes
- Depression
- Withdrawn behavior

Making a Diagnosis

If your doctor suspects that you have a thyroid disorder, he or she will probably investigate the following things.

Your Personal and Family Health History

Thyroid disorders can often be **hereditary**. If you have a family member, like your mother or father, an aunt or uncle, or your brother or sister, who has a thyroid disorder, your chance of developing one may be higher.